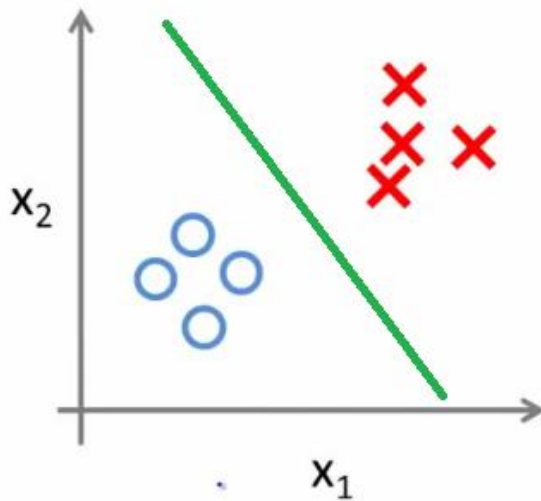
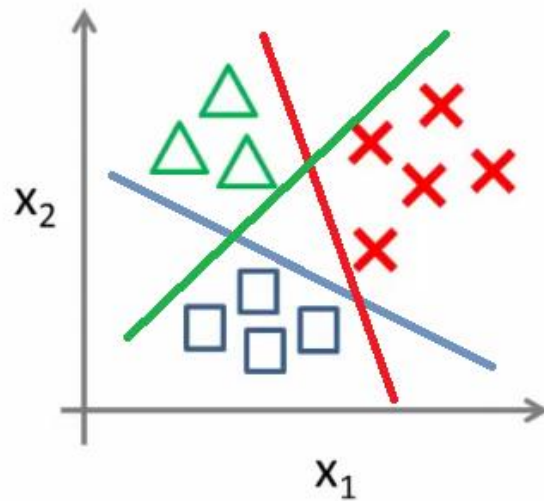


Logistic Regression

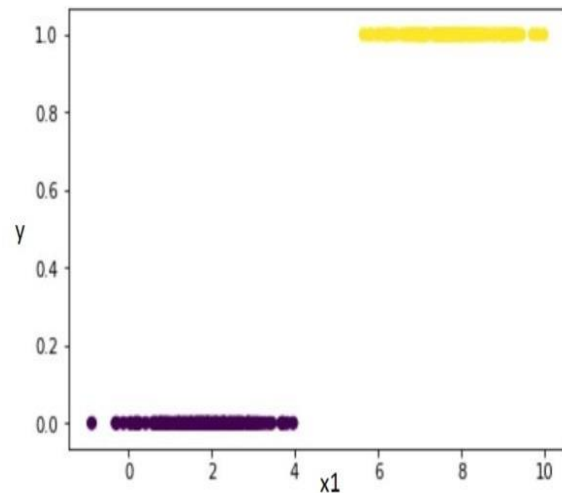
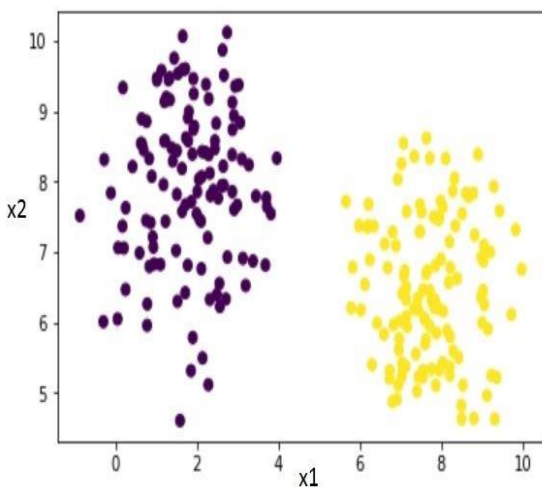
Binary classification:



Multi-class classification:

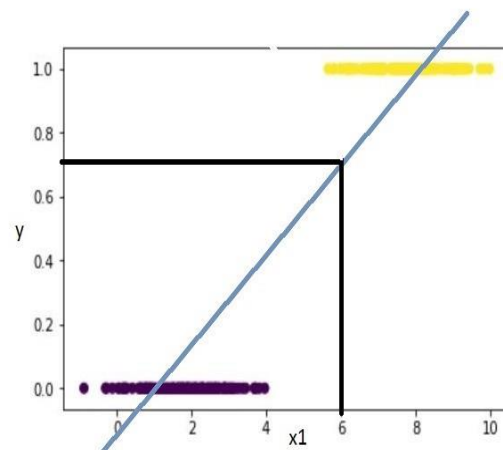
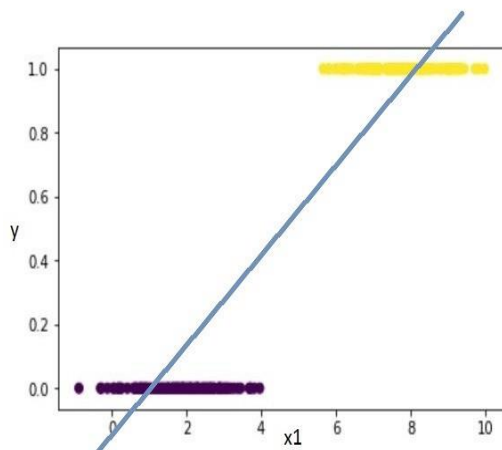


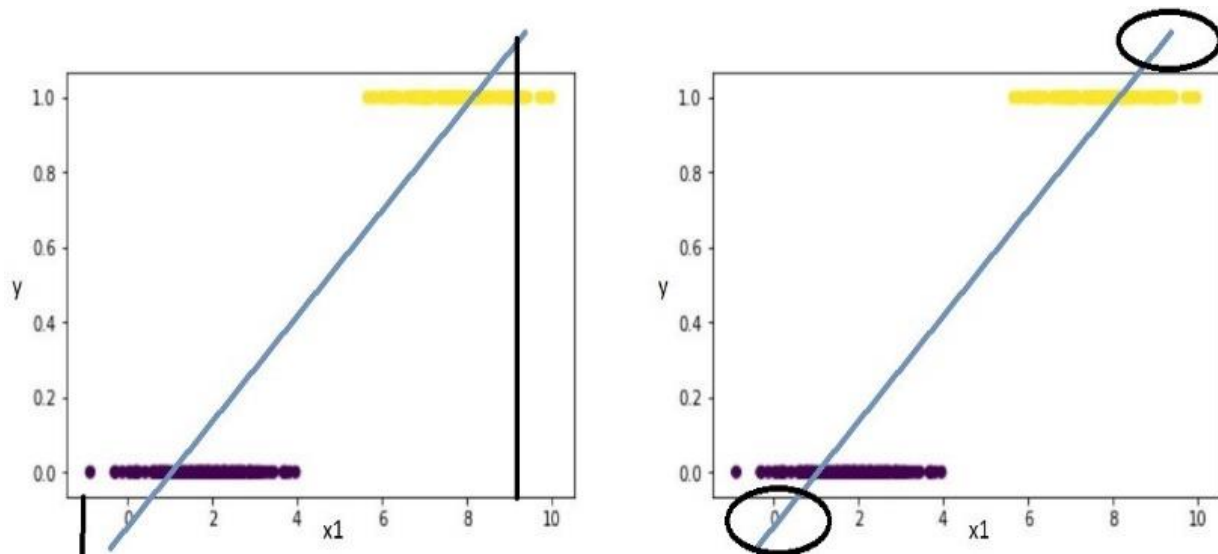
In **binary classification** target variable is either 0 or 1



Given X or (Set of x values) we need to predict whether it's 0 or 1 (Yes/No).

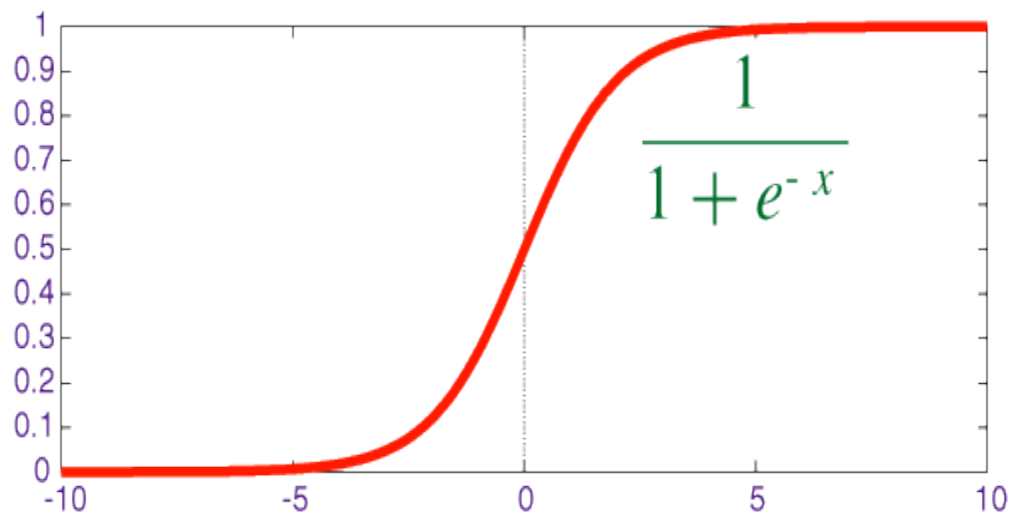
If we apply Linear regression for above data, we get something like this,





We only accept the values between 0 and 1 to decide (Yes/No)

There is an awesome function called **Sigmoid** or **Logistic function**, we use to get the values between 0 and 1

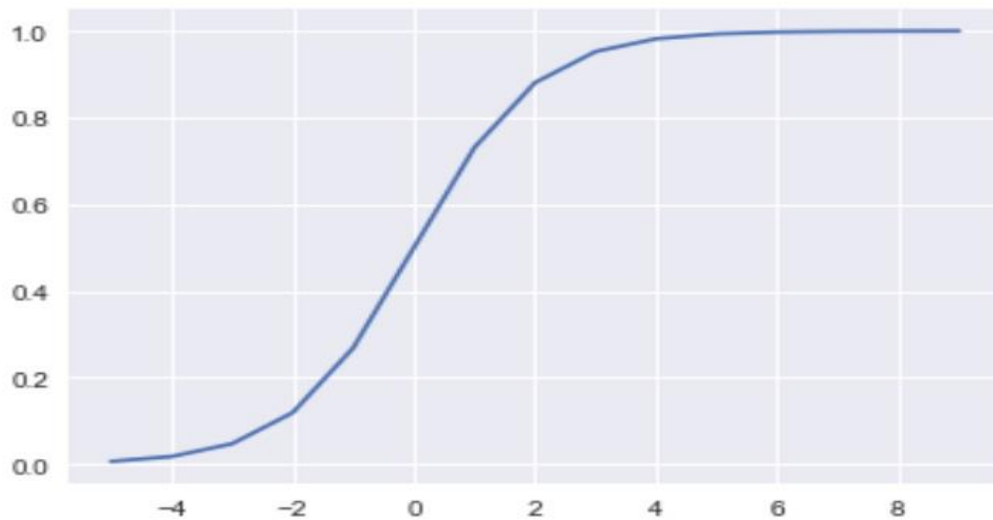


Logistic Regression is a widely used regression model used for classification tasks.

A sigmoid or **logistic** function is a mathematical function having a characteristic “S”-shaped curve or sigmoid curve. Mathematically, it is defined as follows:

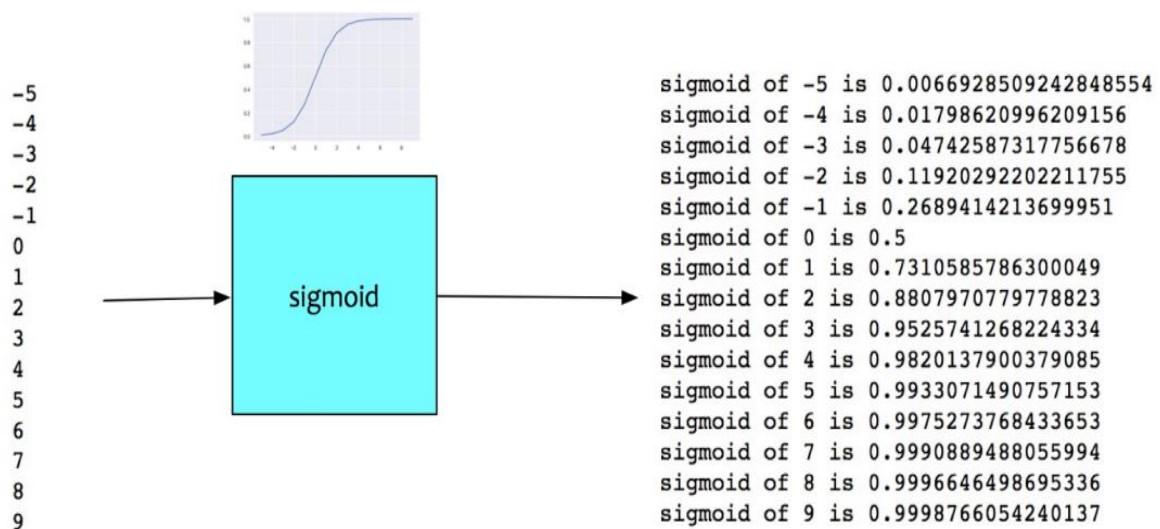
$$\text{sigmoid} = e^y / (1 + e^y)$$

The sigmoid takes the following shape:



It transforms all the values between 0 and 1.

Let us say that we have a set of numbers from -5 to 10. When these set of numbers are transformed using the sigmoid function, all the values are between 0 and 1.



Using sigmoid, any number can be converted to an equivalent probability score.

$$y = \beta_0 + \beta_1 \cdot x$$

$$P(\text{default}) = e^y / (1 + e^y)$$

$$P(\text{default}) = \text{sigmoid}(y)$$